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PEIR: Phase-Energy Integrated Response for Robust Multimodal Remote Sensing Image Matching

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Abstract—Multimodal remote sensing image matching (MRSIM) requires robust feature representations to overcome severe nonlinear radiometric differences (NRDs) and complex geometric distortions. While classical Phase Congruency (PC) methods offer illumination invariance, their normalized responses do not explicitly retain accumulated response magnitude, which suffer from signal response attenuation in low signal-to-noise ratio regions. To complement this property in downstream matching, we propose a novel Phase-Energy Integrated Response (PEIR) framework. Diverging from traditional PC models, PEIR redefines the low-level structural representation by integrating multi-scale phase congruency with local energy responses via Log-Gabor filtering. Building upon this, we formulate an adaptive phase-energy modulated detection mechanism to improve structural saliency under modality-specific noise and severe NRDs. Furthermore, an Orientation-Context Enhancement (OCE) strategy is introduced to aggregate neighboring orientation responses and improve the stability of orientation representation during feature description. Extensive evaluations across five large-scale multimodal datasets demonstrate that PEIR bridges modality gaps, outperforming state-of-the-art handcrafted and deep learning-based approaches, while additional experiments on the planetary OFLP dataset reveal limitations under highly specialized conditions.

Index Terms—Multimodal remote sensing image matching; Nonlinear radiometric differences; Local energy response; Phase-energy integrated response

I. Introduction

In recent years, the rapid development of integrated multi-sensor Earth observation technologies has significantly enhanced our capability to acquire multimodal remote sensing imagery [1]–[3]. Multi-source data—such as optical, infrared, synthetic aperture radar (SAR), and depth images—provide complementary information about surface features, offering immense value for downstream applications including change detection, image fusion, and three-dimensional reconstruction [4]–[7]. Consequently, multimodal remote sensing image matching (MRSIM), which aims to establish accurate correspondences between homologous feature points across different modalities, serves as a fundamental prerequisite for the geometric alignment and synergistic analysis of multi-source data [8]–[11].

Despite its significant value, MRSIM faces more stringent challenges compared to conventional single-modal matching, primarily due to discrepancies in imaging mechanisms, sensor configurations, and observation conditions, which result in pronounced radiometric and geometric differences across modalities [10]. Specifically, these challenges manifest in three aspects. First, different sensors exhibit modality-specific imaging characteristics, leading to inconsistent radiometric responses for the same scene across different modalities, i.e., nonlinear radiometric differences (NRDs), which substantially reduce inter-modal similarity [12]. Second, geometric inconsistencies often arise from differences in projection models and sensor configurations (e.g., projection errors in optical images versus layover effects in SAR images), posing significant difficulties for achieving precise matching. Furthermore, variations in texture richness and modality-specific noise (e.g., speckle noise in SAR) undermine feature stability and induce appearance inconsistencies across multimodal images [10]. Traditional matching methods based on intensity or gradient are highly sensitive to such radiometric and geometric variations, and thus often fail to establish robust correspondences under cross-modal conditions.

To address these formidable challenges, feature-based methods have been widely adopted due to their inherent flexibility and robustness [8]. Among these, Phase Congruency (PC)-based methods have become prevalent [13], [14]. Because PC evaluates the alignment of Fourier components rather than raw intensities, it provides mathematical invariance to signal amplitude, demonstrating strong robustness against illumination changes. However, it formulates the local phase vector upon the normalized sum of even and odd symmetric responses, which discards the accumulated magnitude of local energy. In low signal-to-noise ratio (SNR) or weak-texture multimodal scenarios

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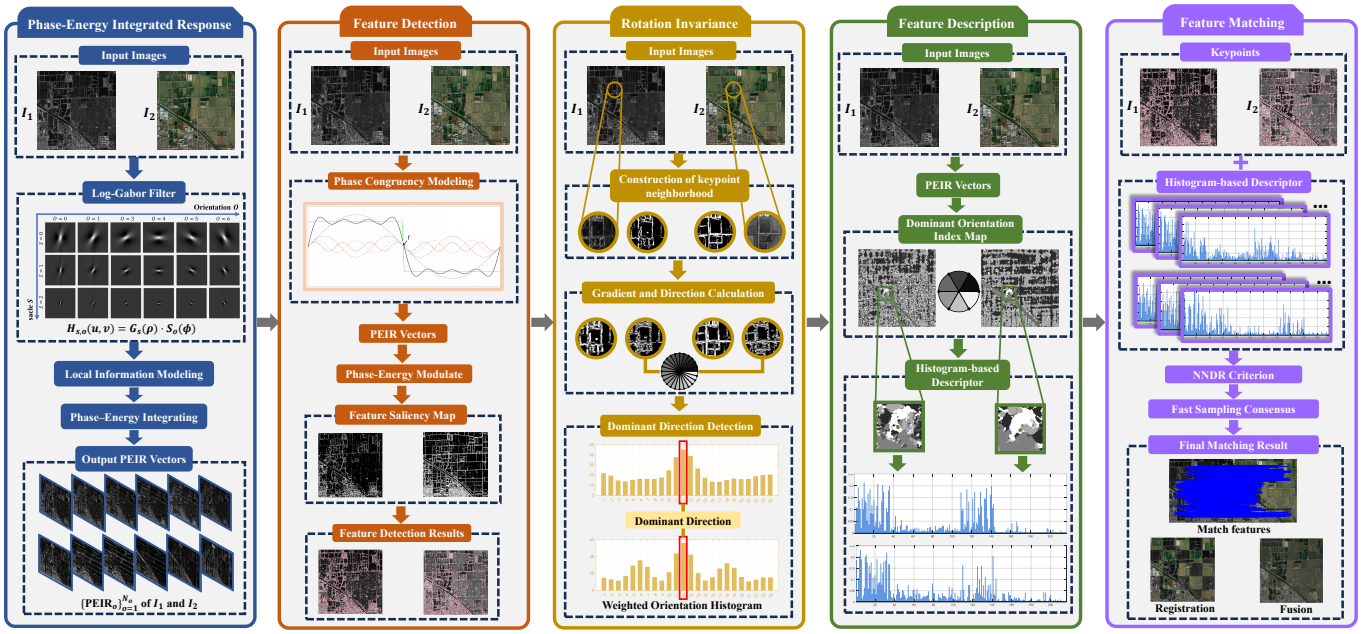

Fig. 1: Overview of PEIR model, which consists of five parts: Phase-Energy Integrated Response, Feature Detection, Implementation of Rotation Invariance, Feature Description and Feature Matching. Inspired by the mechanism of the human visual system for perceiving structural information, PEIR constructs a low-level geometric feature representation that exhibits strong invariance to illumination changes, contrast variations, and NRDs.

(e.g., homogeneous agricultural fields heavily corrupted by SAR speckle noise), this practical property may limit the ability of the standalone normalized PC response to distinguish weak but meaningful structures from unstable responses. From a signal processing perspective, classical PC methods exhibit notable performance degradation under extreme multimodal degradations, primarily due to their inherent sensitivity to noise and the discarding of local energy magnitude information [15], which limits their effective application in low-SNR or strong-speckle scenarios [16]. Furthermore, in geometrically complex scenes, the orientation responses may exhibit ambiguity due to competing orientation components and local fluctuations. Such ambiguity can affect the stability of dominant-direction assignment and reduce descriptor consistency under cross-modal geometric transformations.

To overcome the aforementioned limitations, we propose a Phase-Energy Integrated Response (PEIR) model tailored for multimodal image matching, as shown in Figure 1. Diverging from conventional PC formulations, the proposed PEIR model mathematically recouples phase alignment with multi-scale energy accumulation via multi-orientation Log-Gabor filtering. By introducing a ternary joint response paradigm, PEIR preserves the modality-invariance of phase while compensating for energy deficits in weak structures, establishing a low-level geometric representation that is robust to illumination changes, contrast variations, and severe NRDs. Building upon this foundation, we further formulate an Orientation-Context Enhancement (OCE) strategy. This mechanism aggregates neighboring orientation responses to incorporate local orientation context and improve the stability of descriptor construction.

The main contributions of this paper are outlined below:

- We formulate the Phase-Energy Integrated Response (PEIR) model, a novel signal representation paradigm that integrates phase congruency with multi-scale energy responses and explicitly retains response magnitude for downstream feature processing in low signal-to-noise ratio (SNR) or weak-texture regions.
- We propose phase-energy modulated feature detection mechanism to enhance keypoint repeatability in weak-texture regions, and an Orientation-Context Enhancement (OCE) strategy to improve the stability of orientation representation during feature description.
- We establish a comprehensive evaluation across five distinct multimodal datasets and the planetary OFLP dataset. Experimental results demonstrate that PEIR outperforms state-of-the-art handcrafted and learning-based methods, while revealing limitations relative to domain-specific methods in specialized scenarios.

II. Related work

A. Feature-based Multimodal Image Matching Method

Feature-based methods establish correspondences between homologous keypoints by extracting local invariant feature descriptors from both images. Compared with area-based methods, feature-based methods offer superior robustness to modality-induced radiometric and geometric variations, and typically operate without requiring prior

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georeferenced information, making them more suitable for automated multimodal matching tasks [17]–[20]. The processing pipeline of feature-based methods generally comprises four sequential modules: similarity feature enhancement, feature detection, feature description, and feature matching [10], [21]–[23].

1) Similarity Feature Enhancement: This step is critical for mitigating modality-induced NRDs. Existing methods primarily leverage three types of features: local self-similarity, gradient, and phase congruency. Local self-similarity-based methods, such as DOBSS [24], POS-GIFT [25], OSS [26] and EOSS [27], characterize structures by encoding the internal layout of local image patches. However, their application remains relatively limited due to high computational complexity and sensitivity to imaging noise. Gradient-based methods, such as RIFT [14], MS-HLMO [28] and LNIFT [29], capture structural variations by optimizing gradient computation. While effective, the direct application of traditional gradient operators often fails to capture consistent structures across modalities due to radiometric inconsistencies. PC-based methods, such as HAPCG [30] and WSSF [31], detect salient structures by identifying locations where Fourier components are maximally in phase. Due to its inherent invariance to signal amplitude, PC exhibits strong robustness to illumination changes and geometric distortions, making it well-suited for MRSIM tasks [13]. Recent trends indicate that PC-based features have surpassed local self-similarity and gradient-based methods in popularity due to their superior cross-modal robustness [10].

2) Feature Detection and Description: Feature detection aims to extract stable and repeatable keypoints. Common detectors include Harris, Shi-Tomasi, and FAST, often extended to scale space for multi-scale detection [32]–[34]. For feature description, descriptors must possess rotation and scale invariance. Strategies include rotating the local patch of keypoints according to the dominant orientation or reconstructing the generalized similarity feature map. To achieve scale invariance, methods often construct a scale space or integrate information across multiple scales. However, existing descriptors may suffer from orientation ambiguity in weak-texture regions, leading to inconsistent matching.

3) Feature Matching: The matching process generally consists of descriptor matching and outlier removal. Nearest Neighbor Distance Ratio is widely adopted for initial matching [35]. For outlier removal, geometric consistency-based methods such as Random Sample Consensus (RANSAC) and Fast Sample Consensus (FSC) are employed to fit geometric transformation models and eliminate unreliable correspondences [36]–[38]. Recent methods incorporate additional refinement strategies to further improve matching performance.

B. Learning-based Multimodal Image Matching Method

In addition to handcrafted features, learning-based methods have attracted considerable attention owing to the promising performance of deep learning [39], [40]. These methods can be broadly categorized into four core strategies: (1) replacing key steps in feature-based methods with deep learning techniques; (2) learning to transfer reference and sensed images to the same domain; (3) improve the algorithm performance by implementing data augmentation through image translation [41]; and (4) directly predicting the transformation model between reference and sensed images [42]. Recent advances include detector-free local feature matching with transformers and cross-modal feature matching transformers [43]–[46]. For example, ReDFeat [47] recoupled independent constraints of detection and description of multimodal feature learning with a mutual weighting strategy. XCP-Match [48] is based on a cross-modality completion pre-training strategy, learning cross-modal semantic features through self-supervised pre-training combined with supervised fine-tuning. It effectively overcomes modality discrepancies and geometric distortions without requiring large amounts of annotated data, thereby achieving precise multimodal image matching. CoMatch introduced a covisibility-guided token condenser to dynamically aggregate tokens based on their estimated covisibility scores, a covisibility-assisted attention mechanism to suppress distracting messages from non-covisible regions [49]. While learning-based methods have gained significant popularity, feature-based methods remain widely adopted due to their inherent flexibility and broad applicability. Moreover, feature-based methods offer valuable insights and inductive priors that can support the development of learning-based methods. However, existing learning-based methods often require large amounts of training data and may struggle with generalization across unseen modalities compared to robust handcrafted features designed with physical priors [41], [50].

C. Limitations of Existing Works

Despite the progress made, important limitations still remain. As noted in recent benchmarks, existing datasets often have limitations in terms of modality diversity and ground truth completeness [10], [41], [50], [51]. More importantly, from an algorithmic perspective, current PC-based methods often neglect the cumulative effect of response magnitudes across scales, limiting feature discrimination in some low-contrast regions. Additionally, orientation ambiguity in complex scenes degrades the repeatability of descriptors. The proposed PEIR method addresses these specific limitations by integrating phase and energy information and introducing orientation contrast enhancement, providing a more robust solution for challenging MRSIM tasks.

III. Methodology

A. Phase–Energy Integrated Response Model

In multimodal image matching tasks, source and target images are often acquired through different imaging modalities—such as optical versus SAR, visible versus

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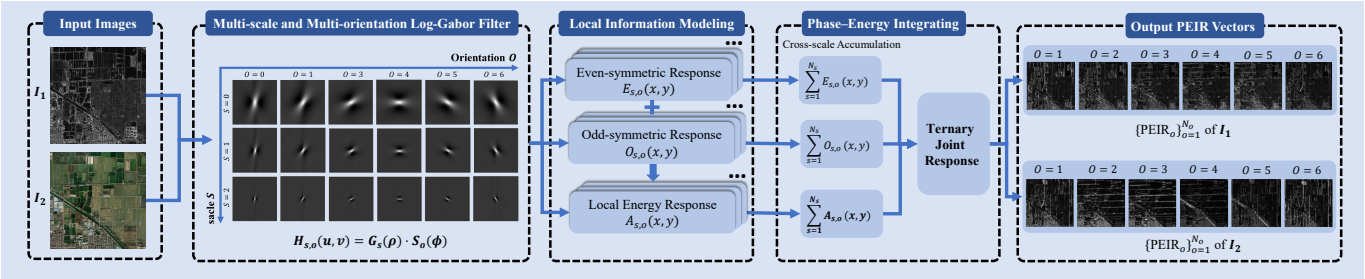

Fig. 2: The framework of the Phase-Energy Integrated Response model, whose input is the multimodal image pair, and the output is a set of PEIR vectors $\{\mathrm{PEIR}_o\}_{o=1}^{N_o}$.

infrared, or multispectral versus panchromatic—leading to significant discrepancies in intensity distributions, texture appearances, and local contrast. Traditional feature descriptors based on gradients or intensities are highly prone to failure in such scenarios due to their sensitivity to NRDs across modalities. To address this challenge, we propose a Phase–Energy Integrated Response (PEIR) model tailored for multimodal image matching, as shown in Figure 2. Inspired by the human visual system's mechanism for perceiving structural information, PEIR constructs a low-level geometric feature representation to improve robustness to illumination changes, contrast variations, and NRDs.

Extensive research on multimodal image matching has shown that the most consistent features across modalities are edges and line-like structures. As illustrated in Figure 7, our experiments further reveal that correctly matched feature points are predominantly concentrated along such edges and line patterns. Salient structures in images, such as edges, corners, and line segments, are not primarily defined by local intensity discontinuities; rather, they are more strongly associated with the phase information of multi-scale, multi-orientation frequency components at specific spatial locations. Because phase information is less sensitive to NRDs under the Fourier transform, phase-based features have cross-modal robustness. The proposed PEIR model is grounded in the theoretical framework of Phase Congruency.

To achieve multi-scale and multi-orientation decomposition of local image structures, PEIR employs Log-Gabor filters as the fundamental analytical tool [52]. Compared to conventional Gabor filters, Log-Gabor filters offer superior control over bandwidth in the frequency domain and avoid the issue of direct current component leakage, making them robust to scale variations. Specifically, the radial component of the Log-Gabor filter is controlled by the relative bandwidth parameter σ_{f} (within the range $0 < \sigma_{\mathrm{f}} < 1$), where a smaller σ_{f} yields narrower bandwidth for higher frequency selectivity, and a larger value enhances scale robustness. In PEIR, we construct a geometrically spaced sequence of wavelengths along the scale dimension as $\lambda_s = \lambda_{\min} \cdot \mathrm{mult}^{s-1}$ for $s = 1, \ldots, N_s$, thereby capturing multi-scale structural information ranging from fine details to global structures.

To introduce orientation selectivity, we define N_o equally spaced orientation angles $\theta_o = (o-1)\pi/N_o$ for $o = 1, \ldots, N_o$, and apply a cosine-shaped angular window to ensure smooth transitions. By multiplying the Fourier transform of the image $\mathcal{F}\{I\}$ with the combined Log-Gabor filter $H_{s,o}$ (radial component times angular window) and applying the inverse Fourier transform, we obtain a complex-valued response:

$$\begin{aligned} \mathrm{EO}_{s,o}(x,y) &= \mathcal{F}^{-1}\left\{\mathcal{F}\{I\} \cdot H_{s,o}\right\} \\ &= E_{s,o}(x,y) + jO_{s,o}(x,y). \end{aligned} \tag{1}$$

where the real part $E_{s,o}$ corresponds to the even-symmetric response, and the imaginary part $O_{s,o}$ corresponds to the odd-symmetric response. This complex representation inherently preserves local phase information, which is crucial for constructing modality-invariant features.

Classical phase congruency methods construct the local phase vector using $\sum_s E_{s,o}$ and $\sum_s O_{s,o}$, and the sum of amplitudes $A_{s,o} = |\mathrm{EO}_{s,o}| = \sqrt{E_{s,o}^2 + O_{s,o}^2}$ is used for normalization, yielding a dimensionless ratio [52], [53]. The amplitude sum is essential to this normalization and contributes to the amplitude- and contrast-invariant properties of PC, but the normalized PC value does not separately retain the accumulated response strength as an explicit downstream feature cue. This omission can lead to response attenuation in low signal-to-noise ratio or weak-texture regions—such as homogeneous agricultural fields in SAR images or shadowed areas in infrared imagery—thereby compromising the repeatability of feature detection. To overcome these limitations, the PEIR model introduces a ternary joint response measure, defined for each orientation o as follows:

$$\begin{aligned} \mathrm{PEIR}(x,y;o) = \Bigg[&\left(\sum\nolimits_{s=1}^{N_s} E_{s,o}(x,y)\right)^2 \\ &+ \left(\sum\nolimits_{s=1}^{N_s} O_{s,o}(x,y)\right)^2 + \left(\sum\nolimits_{s=1}^{N_s} A_{s,o}(x,y)\right)^2 \Bigg]^{1/2} + \epsilon. \end{aligned} \tag{2}$$

where $A_{s,o}$ can be regarded as the local energy response at scale s and orientation o. PEIR explicitly retains the accumulated multi-scale response magnitude alongside the accumulated phase vector, so that response-strength

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information remains available for subsequent detection and description stages.

Let

$$P_o(x,y) = \sqrt{\left(\sum_{s=1}^{N_s} E_{s,o}(x,y)\right)^2 + \left(\sum_{s=1}^{N_s} O_{s,o}(x,y)\right)^2} \quad (3)$$

and $M_o(x,y) = \sum_{s=1}^{N_s} A_{s,o}(x,y)$. Neglecting the small ϵ for clarity, Eq. (2) can be rewritten as $\mathrm{PEIR}(x,y;o) = \sqrt{P_o^2(x,y) + M_o^2(x,y)} = M_o(x,y)\sqrt{1+\kappa_o^2(x,y)}$, where $\kappa_o = P_o/M_o$ and $0 \le \kappa_o \le 1$ by the triangle inequality. Here, M_o preserves the accumulated response magnitude, while κ_o reflects the relative degree of cross-scale phase coherence. Thus, PEIR is essentially an energy-preserving representation modulated by phase coherence.

Finally, the PEIR model outputs a set of orientation-specific response vectors $\{\mathrm{PEIR}_o\}_{o=1}^{N_o}$, which can be directly employed for feature detection and description. Compared to conventional PC, PEIR significantly enhances robustness to multimodal imaging degradations—such as speckle noise, blur, and low contrast—while preserving the modality-invariant properties of phase information.

B. Feature Detection

In multimodal image matching, reliable feature detection not only demands high repeatability but also requires that the extracted keypoints correspond to geometric structures with shared semantic meaning across modalities. To this end, building upon the PEIR model, we propose a phase-energy modulated feature detection mechanism that jointly leverages the structural discriminability of PC and the energy-enhanced characteristics of PEIR, thereby generating feature points robust to NRDs. The framework of feature detection is shown in Figure 3.

The PC vector $\mathrm{PC}(x,y;o)$ is defined as the normalized degree of energy concentration at (x,y) in orientation o, resulting from the alignment of local frequency components in phase. Its classical formulation can be expressed as:

$$\mathrm{PC}(x,y;o) = \frac{\sum_{s=1}^{N_s} A_{s,o}(x,y)\left[\cos(\Delta\phi_{s,o}) - |\sin(\Delta\phi_{s,o})|\right]}{\sum_{s=1}^{N_s} A_{s,o}(x,y) + \epsilon}. \quad (4)$$

where $A_{s,o}$ denotes the magnitude of the filter response at scale s and orientation o, and $\Delta\phi_{s,o}$ is the angular deviation of the phase at that scale from the multi-scale average phase vector. $\epsilon = 10^{-4}$ is a small constant added for numerical stability. The numerator quantifies the degree of phase alignment across scales: when all components are in phase (i.e., $\Delta\phi_{s,o} \approx 0$), the term $\cos(\Delta\phi) - |\sin(\Delta\phi)|$ approaches 1, maximizing the PC value; conversely, if phases are randomly distributed, this term tends toward zero. Consequently, PC suppresses spurious responses caused by modality-specific textures or noise, while highlighting regions with salient geometric structures.

However, in multimodal remote sensing scenarios, certain genuine structures—such as smooth roads or low-contrast boundaries—may exhibit weak PC responses due to low signal-to-noise ratios, causing them to be discarded during subsequent non-maximum suppression. To mitigate this issue, we introduce the PEIR as an energy-based modulation factor to adaptively enhance the original PC map.

Specifically, for each orientation o, the PC_o is first multiplied element-wise by the corresponding PEIR_o to obtain a phase-energy modulated local saliency map:

$$\widetilde{\mathrm{PC}}_o(x,y) = \mathrm{PC}_o(x,y) \cdot \mathrm{PEIR}_o(x,y). \quad (5)$$

Eq. (5) has a clear physical interpretation: at locations where phase alignment is strong (high PC) and substantial multi-scale response magnitude is retained in PEIR, the saliency is amplified. Conversely, when phase consistency is weak, the multiplication by PC suppresses the response even if the corresponding PEIR magnitude is large, while PEIR retains response-strength information for the subsequent detection. Thus, PC provides phase-consistency selectivity, whereas PEIR contributes complementary response-magnitude information. Subsequently, the modulated responses across all orientations are weighted and fused, and normalized by the total PEIR to yield the final unified feature saliency map:

$$\mathcal{S}(x,y) = \frac{\sum_{o=1}^{N_o} \widetilde{\mathrm{PC}}_o(x,y)}{\sum_{o=1}^{N_o} \mathrm{PEIR}_o(x,y) + \epsilon}. \quad (6)$$

The saliency map S has three key advantages: 1) Geometric semantic consistency: Governed by PC, it ensures that detected responses correspond to structures shared across modalities; 2) Enhancement of weak structures: Modulated by PEIR, it improves the visibility of genuine features under low contrast or noise interference; 3) Multi-orientation information fusion: By aggregating responses across orientations, it eliminates orientation dependency and establishes a rotation-robust foundation for feature detection.

Based on S, feature points are extracted using local maximum detection strategy (e.g., FAST [34]) and ranked according to their saliency strength, with the top N most stable points retained as the final detection output. Since S is inherently highly sensitive to modality-invariant geometric structures, the detected keypoints exhibit excellent cross-modal repeatability across multimodal image pairs, thereby providing a high-quality set of feature points for subsequent description and matching stages.

C. Implementation of Rotation Invariance

To achieve rotation invariance, we assign a dominant direction to each detected keypoint based on the saliency map S as shown in Figure 4, to reduce the sensitivity of the subsequent descriptor to rotation.

Direction assignment is based on the statistical characteristics of gradient distributions in the neighborhood of keypoint. For each keypoint $\mathbf{p} = (x,y)$, a circular

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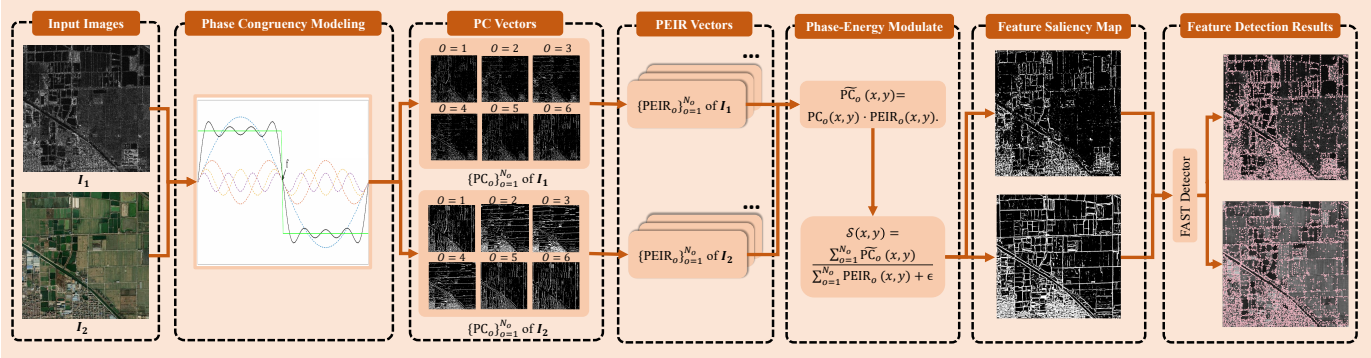

Fig. 3: The framework of feature detection. Building upon the PEIR model, this feature detector jointly leverages the structural discriminability of PC and the energy-enhanced characteristics of PEIR, thereby generating feature points robust to NRDs.

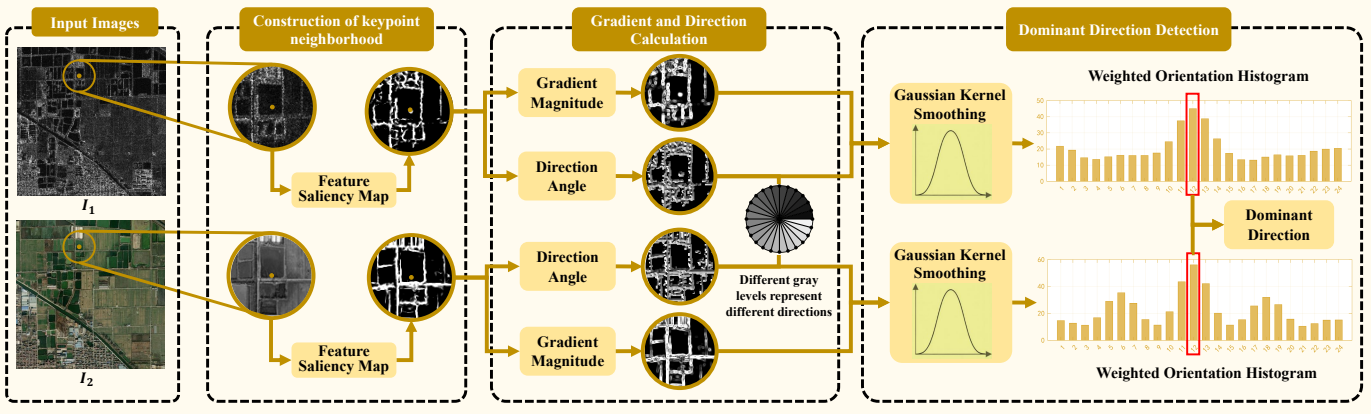

Fig. 4: The framework of implementation of rotation invariance. It assigns a dominant direction to each detected keypoint based on the saliency map S, ensuring that the subsequent descriptor remains invariant under different rotation angles.

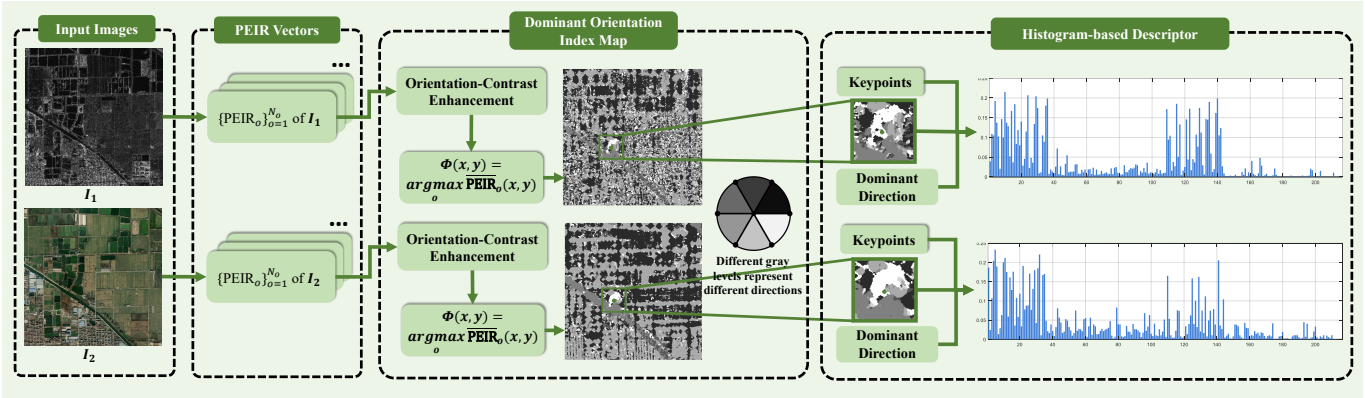

Fig. 5: The framework of feature description. The descriptor directly takes the PEIR as its underlying feature input, fully exploiting PEIR's high sensitivity to geometric structures and its modality invariance in cross-modal scenarios.

neighborhood with a radius of R is constructed. Within this region, we first compute the gradient magnitude and direction angle for each pixel:

$$G(\mathbf{u}) = \sqrt{\left(\frac{\partial I}{\partial x}(\mathbf{u})\right)^2 + \left(\frac{\partial I}{\partial y}(\mathbf{u})\right)^2},$$
$$\theta(\mathbf{u}) = \operatorname{atan2}\left(\frac{\partial I}{\partial y}(\mathbf{u}), \frac{\partial I}{\partial x}(\mathbf{u})\right) \in [0°, 360°). \tag{7}$$

where $\mathbf{u}$ denotes the coordinates of an arbitrary pixel

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within the neighborhood. The gradient direction of the saliency map S is typically determined by salient geometric structures and exhibits cross-modal consistency.

Subsequently, the interval $[0°, 360°)$ is uniformly divided into N direction bins (we set $N = 24$), and a weighted direction histogram $H[\cdot]$ is constructed, where the accumulated value in each bin is the sum of gradient magnitudes of all pixels falling within that angular interval:

$$H[k] = \sum_{\mathbf{u}\in\Omega_k} G(\mathbf{u}), \quad k = 0, 1, \ldots, N-1. \tag{8}$$

where Ω_k denotes the set of pixels falling into the k-th direction bin. The histogram is smoothed with a Gaussian kernel. Let $H_{\max}$ be the highest peak. All local maxima satisfying $H[k] \geq \eta \cdot H_{\max}$ (with directional peak ratio $\eta = 0.8$ in PEIR) are selected as dominant directions $-\phi \in [0°, 360°)$. During description, the local sampling grid is rotated by $-\phi$ to align the descriptor with the dominant orientation, ensuring rotation invariance.

D. Feature Description

After keypoint detection and dominant direction assignment, we construct the local descriptor based on multi-orientation PEIR to achieve discriminative encoding of geometric structures in multimodal images. As shown in Figure 5, the descriptor directly takes the PEIR as its underlying feature input, fully exploiting PEIR's high sensitivity to geometric structures and its modality invariance in cross-modal scenarios.

To achieve the robust representation of geometric structures in multimodal images, we adopt a local histogram-based descriptor indexed by dominant orientations. The core idea of this strategy is to encode local structural semantics into discrete orientation categories and construct a discriminative feature vector by aggregating their spatial distribution statistics. Specifically, given a set of PEIR maps $\{\mathrm{PEIR}_o(x,y)\}_{o=1}^{N_o}$, we first determine, at each pixel location, the orientation channel with the strongest response, yielding a dominant orientation index map: $\Phi(x,y) = \arg\max_o \mathrm{PEIR}_o(x,y)$. Subsequently, the neighborhood around the keypoint is partitioned into sub-regions, and a histogram of orientation index $\Phi(x,y)$ is computed within each sub-region. These histograms are then concatenated and normalized to form the final descriptor.

By encoding local structures into discrete dominant orientation indices, the descriptor captures essential geometric features such as edges and stripes, thereby achieving semantic representation that aligns closely with physical structures. By combining phase-related structural cues with response-magnitude information, the extracted dominant orientation of PEIR can exhibit improved consistency across multimodal image pairs under illumination variations, contrast differences, and NRDs, supporting more reliable cross-modal matching. Moreover, the method uses histogram statistics of orientation indices rather than high-dimensional floating-point features, providing a compact descriptor representation for multimodal remote sensing image matching and enhancing computational efficiency.

However, in practical multimodal scenarios, orientation-specific PEIR responses may contain competing peaks or isolated fluctuations, especially in weak-texture regions and scenes with intersecting structures. Such variations can affect the stability of the per-pixel dominant-orientation index and reduce descriptor consistency across modalities. To address this problem, we introduce an Orientation-Context Enhancement (OCE) mechanism, designed to incorporate local orientation context and reduce the sensitivity of orientation selection to isolated response fluctuations. OCE is applied directly to the set of PEIR via a one-dimensional weighted convolution along the orientation axis: at each spatial location, the response of current orientation is fused with those of its immediate neighbors using weights $[0.5, 1, 0.5]$:

$$\begin{aligned}\widetilde{\mathrm{PEIR}}_o(x,y) = {} & 0.5\,\mathrm{PEIR}_{o-1}(x,y) \\ & + \mathrm{PEIR}_o(x,y) + 0.5\,\mathrm{PEIR}_{o+1}(x,y).\end{aligned} \tag{9}$$

For the first orientation (o=1) and the last orientation (o=No), we adopt a circular (periodic) boundary condition. For $o = 1$: $\widetilde{\mathrm{PEIR}}_1 = 0.5\,\mathrm{PEIR}_{N_o} + \mathrm{PEIR}_1 + 0.5\,\mathrm{PEIR}_2$; For $o = N_o$: $\widetilde{\mathrm{PEIR}}_{N_o} = 0.5\,\mathrm{PEIR}_{N_o-1} + \mathrm{PEIR}_{N_o} + 0.5\,\mathrm{PEIR}_1$.

The OCE mechanism performs local aggregation along the orientation dimension by combining the current response with its two adjacent orientation channels in Eq. (9). This operation incorporates local orientation context and reduces the sensitivity of the subsequent dominant-orientation assignment to isolated response fluctuations. The response distribution after OCE is therefore expected to be smoother. Subsequently, all aggregated orientation responses are normalized by their L2 norm:

$$\overline{\mathrm{PEIR}}_o(x,y) = \frac{\widetilde{\mathrm{PEIR}}_o(x,y)}{\sqrt{\sum_{k=1}^{N_o} \widetilde{\mathrm{PEIR}}_k^2(x,y) + \epsilon}}. \tag{10}$$

Finally, the dominant orientation index map is determined from the normalized PEIR:

$$\Phi(x,y) = \arg\max_o \overline{\mathrm{PEIR}}_o(x,y). \tag{11}$$

By incorporating local orientation context, the OCE mechanism provides a smoother orientation representation for constructing the map Φ. The histogram-based descriptor built upon Φ preserves the geometric semantic advantages of the original method while improving the robustness of orientation representation under noise, weak textures, and complex structures.

E. Feature Matching

We adopt a two-stage strategy to achieve high-precision feature matching, as shown in Figure 6: first, an initial matching is performed based on descriptor similarity, followed by outlier rejection using geometric consistency constraints. The initial matching employs the Nearest

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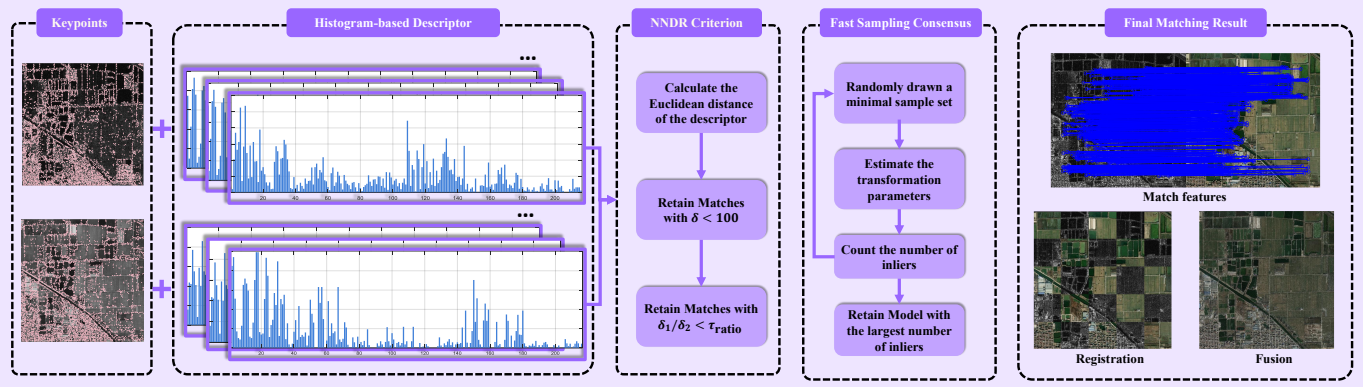

Fig. 6: The framework of feature matching, which adopts a two-stage strategy to achieve high-precision feature matching.

Neighbor Distance Ratio (NNDR) criterion with a distance ratio threshold $\tau_{ratio} = 1$, allowing all candidate matches to proceed to geometric verification. To reduce false matches caused by NRDs, structural ambiguities, or noise, we use the Fast Sampling Consensus (FSC) algorithm [37] with a reprojection error threshold of 30 pixels.

IV. Experimental Results

A. Experimental settings

1) Dataset Description: To evaluate the proposed PEIR method, we have collected five distinct multimodal remote sensing image datasets, each presenting unique challenges and covering a variety of remote sensing scenarios. The detailed descriptions of the datasets are as follows:

MMIM Dataset [42]: This dataset is a comprehensive database designed to establish a unified standard for experimental evaluation, encompassing multimodal image pairs from medical, remote sensing, and computer vision domains. It consists of 18 modality pairs: in the Medical Research category, it includes cross-matched MR T1, T2, and PD-weighted images, MRI-PET, SPECT-CT, and retinal images; in Remote Sensing Research, it contains UAV cross-seasonal images, optical day-night pairs, LiDAR depth-optical pairs, IR-optical pairs, MAP-optical pairs, optical time-series pairs, and SAR-optical pairs; and in Computer Vision, it features visible-infrared image pairs, visible-near-infrared pairs, cross-seasonal visible pairs, day-night image pairs, and image-painting pairs. In the experiment, we only used the remote sensing images from this dataset for evaluation. This dataset provides manually annotated ground-truth (GT) homography transformation matrices for each image pair.

HOWP Dataset [54]: This dataset comprises 50 sets of multimodal image pairs, including depth optical, infrared optical, map optical, SAR optical, and nighttime images. The image pairs exhibit not only differences in noise levels and contrast but also geometric transformations, such as scale, rotation, and displacement. The dataset sources include airborne LiDAR data for depth images, Landsat TM-5 or airborne infrared sensors for infrared images, Google Maps for maps, GF3 satellite data for SAR, and nighttime imagery from various satellites. The image sizes range from 381 to 750 pixels. This dataset provides GT homography transformation matrices for each image pair.

Multi-Resolution-SAR Dataset [55]: This dataset contains 1000 optical and SAR image pairs, covering data from four commercial satellites with resolutions ranging from 0.16 m to 10 m. It includes SAR data from Sentinel-1, GF-3, HT1-A, and Umbra satellites with multiple polarization modes. The dataset features six different scene types, such as urban, rural, plains, hills, mountains, and water bodies, with spatial resolutions ranging from 0.16m to 10m, collected from multiple countries and cities. This dataset provides the GT homography transformation matrix for optical–SAR image pairs.

MapData Dataset [56]: MapData is a global, diversified dataset spanning 233 geographical sampling points, with original images ranging from 7000×5000 to 20000×15000 pixels. After thorough cleaning, the dataset provides 1910 image pairs of electronic maps and visible images (512×512 pixels), with a mix of manual and automated ground truth. The sampling points cover a variety of typical scenes, including urban built-up areas, rural settlements, mountainous terrain, plains, basins, and desert regions. The original images have spatial resolutions ranging from 3m to 500m.This dataset provides GT homography transformation matrices for each image pair.

SRIF Dataset [57]: This dataset contains multimodal image pairs, including optical-optical, optical-infrared, optical-SAR, optical-depth, optical-map, and nighttime images. The author has applied random geometric transformations, such as rotations ($\in[0°, 90°]$) and scaling factors ($\in[0.5, 2]$), to the target images to generate this dataset. The SRIF dataset consists of 1,200 image pairs, covering six typical multimodal image types. This dataset provides GT homography transformation matrices for each image pair.

2) Experimental details: The comparison includes five handcrafted feature-based methods: WSSF [31], EOSS [27], LNIFT [29], SRIF [57], and POS-GIFT [25], as well

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TABLE I: The quantitative comparison results on MMIM datasets. The best and second-best results in each category are highlighted in Bold and Underline, respectively.

Methods	Metrics	SAR-Optical	Cross Season	Day-Night	Depth-Optical	Infrared-Optical	Map-Optical	Optical-Optical
WSSF	NCM	79.2	90.4	204.8	61.8	253.0	246.3	338.0
	CMR(%)	71.22	46.62	73.70	81.27	82.88	92.72	67.50
	MSE(↓)	3.514	5.219	4.298	3.247	3.184	3.288	3.072
	SR(%)	87.80						
EOSS	NCM	241.8	581.0	262.1	683.5	227.1	820.1	463.5
	CMR(%)	43.99	48.91	42.86	17.97	20.31	10.61	53.23
	MSE(↓)	6.667	6.679	5.719	4.640	3.708	9.082	6.482
	SR(%)	46.34						
LNIFT	NCM	53.3	99.4	38.0	172.1	53.5	403.3	615.0
	CMR(%)	37.64	59.25	39.66	87.50	49.17	63.91	53.22
	MSE(↓)	7.038	4.948	7.652	2.976	2.008	5.399	6.207
	SR(%)	75.61						
SRIF	NCM	146.0	305.0	121.4	147.9	69.5	89.0	318.9
	CMR(%)	6.07	13.47	4.64	6.16	4.79	4.10	13.04
	MSE(↓)	5.587	5.973	6.763	3.941	2.913	7.290	6.115
	SR(%)	65.85						
POS-GIFT	NCM	758.0	49.2	52.1	251.1	406.5	494.6	104.5
	CMR(%)	66.24	45.79	60.00	25.00	25.00	14.29	66.67
	MSE(↓)	5.728	6.569	5.156	4.531	3.604	8.884	4.496
	SR(%)	46.34						
XoFTR	NCM	207.2	155.0	286.4	474.3	544.3	37.0	542.5
	CMR(%)	88.48	29.37	73.62	86.28	87.63	46.69	69.62
	MSE(↓)	4.218	6.713	3.878	3.465	1.748	6.389	2.614
	SR(%)	75.61						
EDM	NCM	522.3	601.4	133.6	471.6	399.8	49.1	827.2
	CMR(%)	62.47	20.00	65.32	46.92	40.16	38.89	50.00
	MSE(↓)	2.752	6.097	2.926	2.832	2.599	3.911	2.225
	SR(%)	51.22						
ReDFeat	NCM	103.3	72.4	62.8	308.0	208.3	248.0	202.2
	CMR(%)	69.65	45.33	77.28	80.12	87.17	79.18	79.14
	MSE(↓)	3.270	6.779	2.981	2.903	2.713	3.245	3.318
	SR(%)	97.56						
PEIR	NCM	957.8	398.2	603.8	573.2	632.3	843.0	995.8
	CMR(%)	74.45	64.47	85.39	96.76	95.29	95.90	85.92
	MSE(↓)	2.502	4.482	3.559	2.851	2.517	2.436	2.558
	SR(%)	97.56						
Number of image pairs		6	5	5	8	4	7	6

as three learning-based feature methods: XoFTR [44], EDM [45], and RedFeat [47]. The handcrafted feature-based algorithms, including PEIR, were implemented in MATLAB 2023a and executed on an Intel Core i9-13900HX CPU, while the learning-based methods were implemented in Python and executed on an NVIDIA RTX 4090 GPU. All experiments were conducted on the same workstation; however, due to the use of different equipment and software environments, it was impossible to ensure identical hardware conditions, so only a loose comparison of the execution times for the various methods could be made. The handcrafted algorithms were run with the default settings as described in their respective papers, and the parameters for the learning-based methods were also kept at their default values.

For the evaluation of matching performance on the MMIM, HOWP, and SRIF datasets, four primary metrics were utilized: Number of Correct Matches (NCM), Root Mean Square Error (MSE), Success Rate (SR), and Correct Matching Rate (CMR). These metrics are designed to assess both the accuracy and robustness of feature matching across different image pairs. Specifically, the NCM serves as the primary evaluation criterion for image matching, where correspondences with residuals below 3 pixels are considered correct matches. MSE calculates the average deviation between the matched points and the ground truth correspondences, providing a direct measure of the overall matching accuracy. CMR represents the proportion of matches that are deemed correct before outlier removal, serving as an indicator of the method's precision in identifying valid correspondences. The Success Rate (SR) is defined as the percentage of successfully matched image pairs (i.e., those whose MSE is below a threshold of 5 pixels) out of the total number of tested

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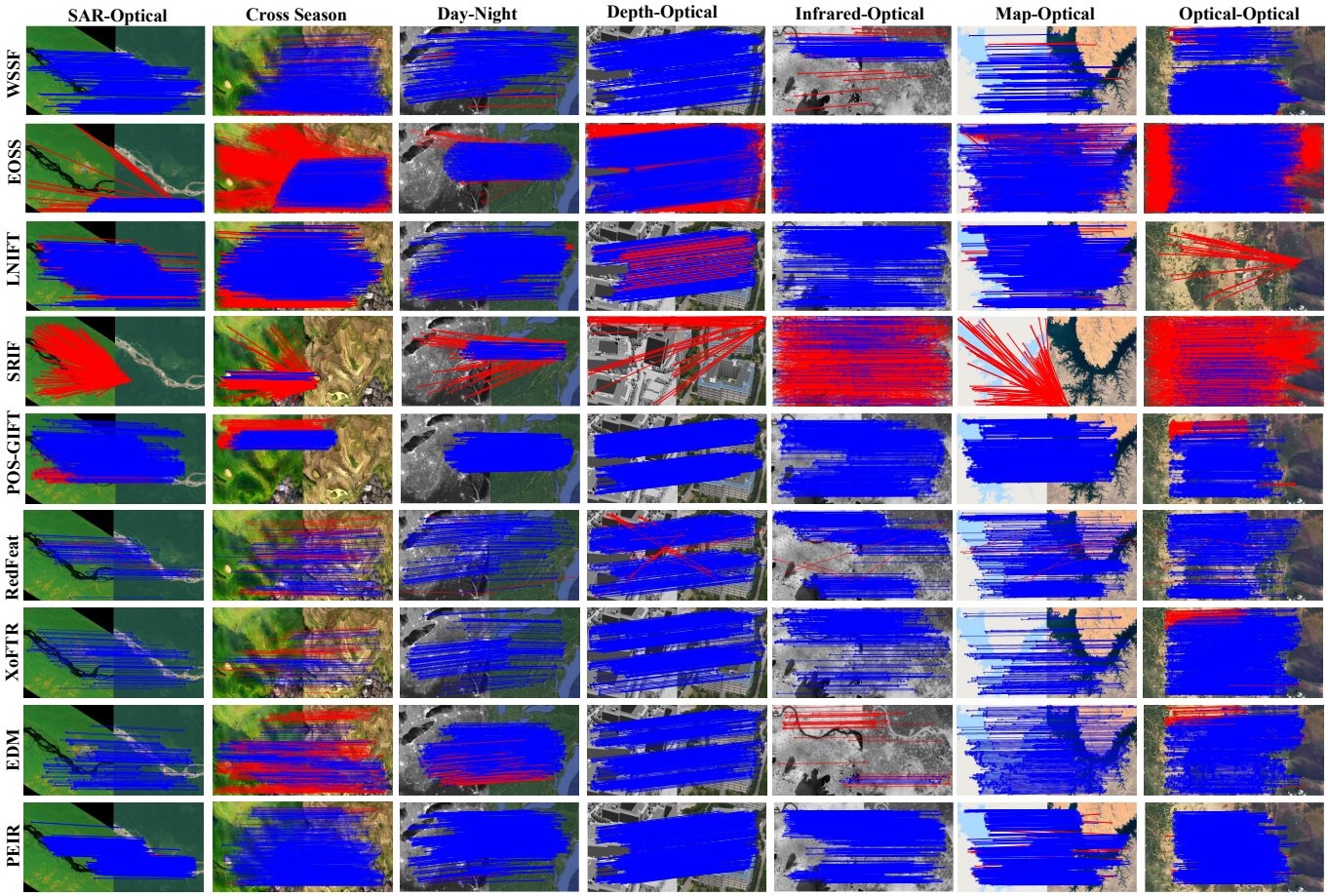

Fig. 7: Qualitative comparison results of all tested methods for seven typical image pairs in MMIM dataset. Blue and red lines indicate correct matches and mismatches, respectively.

image pairs, which is commonly used to evaluate the overall robustness and reliability of a matching algorithm under varying conditions.

In the case of Homography Evaluation for the Multi-Resolution-SAR and MapData datasets, we focus on the 4-point MSE, which computes the average projection error for the four corner points after applying the homography transformation. To provide a more comprehensive evaluation, we report the Area Under the Curve (AUC) at three different error thresholds—3 px, 5 px, and 10 px—to assess the algorithm's performance under varying degrees of geometric distortion.

B. Parameter Settings

In the phase–energy integrated response model, we set the number of scales to $N_s = 4$, the number of orientations to $N_o = 6$, and the relative bandwidth parameter to $\sigma_{\mathrm{f}} = 0.75$. In the implementation of rotation invariance, we set the radius of the circular neighborhood to $R = 96$, the number of direction bins to $N = 24$, and the directional peak ratio to $\eta = 0.8$. In the feature matching, we set the distance ratio threshold to $\tau_{\mathrm{ratio}} = 1$ and the reprojection error threshold to 30 pixels.

C. Result on the MMIM Dataset

Table I and Figure 7 present the qualitative and quantitative matching results across seven representative modality pairs in the MMIM dataset. As shown in Table 1, some conventional handcrafted methods exhibit pronounced limitations when confronted with modality-specific degradations. Specifically, WSSF and LNIFT achieve notably low NCM values under the SAR–Optical modality, indicating their insufficient robustness to the strong speckle noise and geometric distortions inherent in SAR imagery. SRIF shows a pronounced performance drop in the Infrared–Optical modality, primarily because it is not sufficiently sensitive to the significant spectral discrepancies between infrared and optical images. In contrast, the SR performance of EOSS and POS-GIFT is markedly inferior, with a success rate of only 46.34%, suggesting that these methods often fail to produce stable and reliable matching results. Moreover, EDM employs a deeper yet parameter-efficient CNN architecture for multi-level feature extraction, prioritizing matching efficiency at the cost of reduced accuracy. As more recently developed deep-learning cross-modal matching methods, XoFTR and ReDFeat are able to obtain a certain number of correct correspondences, achieving SR values of 75.61% and

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TABLE II: The quantitative comparison results on HOWP datasets. The best and second-best results in each category are highlighted in Bold and Underline, respectively.

Methods	Metrics	1DSMsets	2InfreadSets	3MapSets	4SARSets	5NightSets
WSSF	NCM	18.8	100.3	172.7	16.6	43.4
	CMR(%)	58.21	70.63	68.65	40.22	55.84
	MSE(↓)	3.554	3.775	3.022	6.785	4.325
	SR(%)			72.00		
EOSS	NCM	492.3	432.5	265.3	442.9	360.3
	CMR(%)	68.18	39.12	41.64	49.39	24.96
	MSE(↓)	5.794	6.531	6.691	6.655	7.754
	SR(%)			66.00		
LNIFT	NCM	104.6	272.7	179.6	32.1	160.0
	CMR(%)	14.01	34.39	51.56	5.30	32.18
	MSE(↓)	7.319	7.071	5.921	8.772	7.086
	SR(%)			48.00		
SRIF	NCM	89.5	113.4	262.7	100.5	103.6
	CMR(%)	5.40	5.67	15.80	4.54	4.63
	MSE(↓)	5.385	6.171	4.917	6.284	6.024
	SR(%)			64.00		
POS-GIFT	NCM	95.8	108.4	136.7	83.5	117.2
	CMR(%)	41.36	56.82	61.73	30.73	29.46
	MSE(↓)	6.821	6.134	5.872	7.105	7.436
	SR(%)			52.00		
XoFTR	NCM	32.5	862.8	276.4	186.4	73.2
	CMR(%)	56.41	75.14	63.00	38.30	48.96
	MSE(↓)	6.536	1.428	5.353	5.462	6.274
	SR(%)			88.00		
EDM	NCM	379.5	745.8	260.2	184.8	251.5
	CMR(%)	79.91	68.02	72.27	49.15	42.54
	MSE(↓)	4.976	4.547	5.734	5.745	5.052
	SR(%)			74.00		
ReDFeat	NCM	159.9	184.0	131.1	59.5	25.2
	CMR(%)	65.59	77.05	62.41	56.93	49.02
	MSE(↓)	5.578	3.128	6.071	4.823	5.374
	SR(%)			88.00		
PEIR	NCM	170.1	338.7	424.6	218.0	198.8
	CMR(%)	94.71	80.00	75.00	63.34	49.70
	MSE(↓)	3.784	2.949	2.753	4.101	3.247
	SR(%)			90.00		
Number of image pairs		10	10	10	10	10

97.56%, respectively. However, their overall performance still falls short of the proposed PEIR method. By comparison, the proposed PEIR ranks first or second across nearly all metrics and modalities. It delivers the highest NCM in five categories (e.g., 957.8 for SAR-Optical and 995.8 for Optical-Optical) while matching the best overall SR of 97.56%. Moreover, PEIR achieves superior CMR (up to 96.76% in Depth-Optical) and the lowest MSE in most cases (e.g., 2.436 for Map-Optical), demonstrating exceptional precision and robustness.

D. Result on the HOWP Dataset

Table II and Figure 8 present the qualitative and quantitative matching results on the HOWP dataset. As detailed in Table II, LNIFT, SRIF, and POS-GIFT achieve notably low success rates (SR) of 48.00%, 64.00%, and 52.00%, respectively, with particularly poor performance in the DSMsets and SARSets scenarios (e.g., NCM as low as 18.80 for WSSF in DSMsets and CMR of 30.73% for POS-GIFT in SARSets). These shortcomings arise primarily from their sensitivity to complex geometric structures and severe NRDs, resulting in insufficient keypoint repeatability and unstable descriptors. In comparison, learning-based methods such as XoFTR, EDM, and ReDFeat generally outperform the handcrafted baselines, benefiting from their data-driven feature representations. XoFTR attains a strong SR of 88.00% with high NCM in the InfraredSets, while EDM and ReDFeat reach SR values of 74.00% and 88.00%, respectively. However, these methods still introduce a considerable number of incorrect correspondences, as evidenced by their relatively lower CMR and higher MSE in several modalities (e.g., EDM's CMR drops to 42.54% in NightSets with MSE of 5.052 pixels). This indicates that, despite improved generalization in

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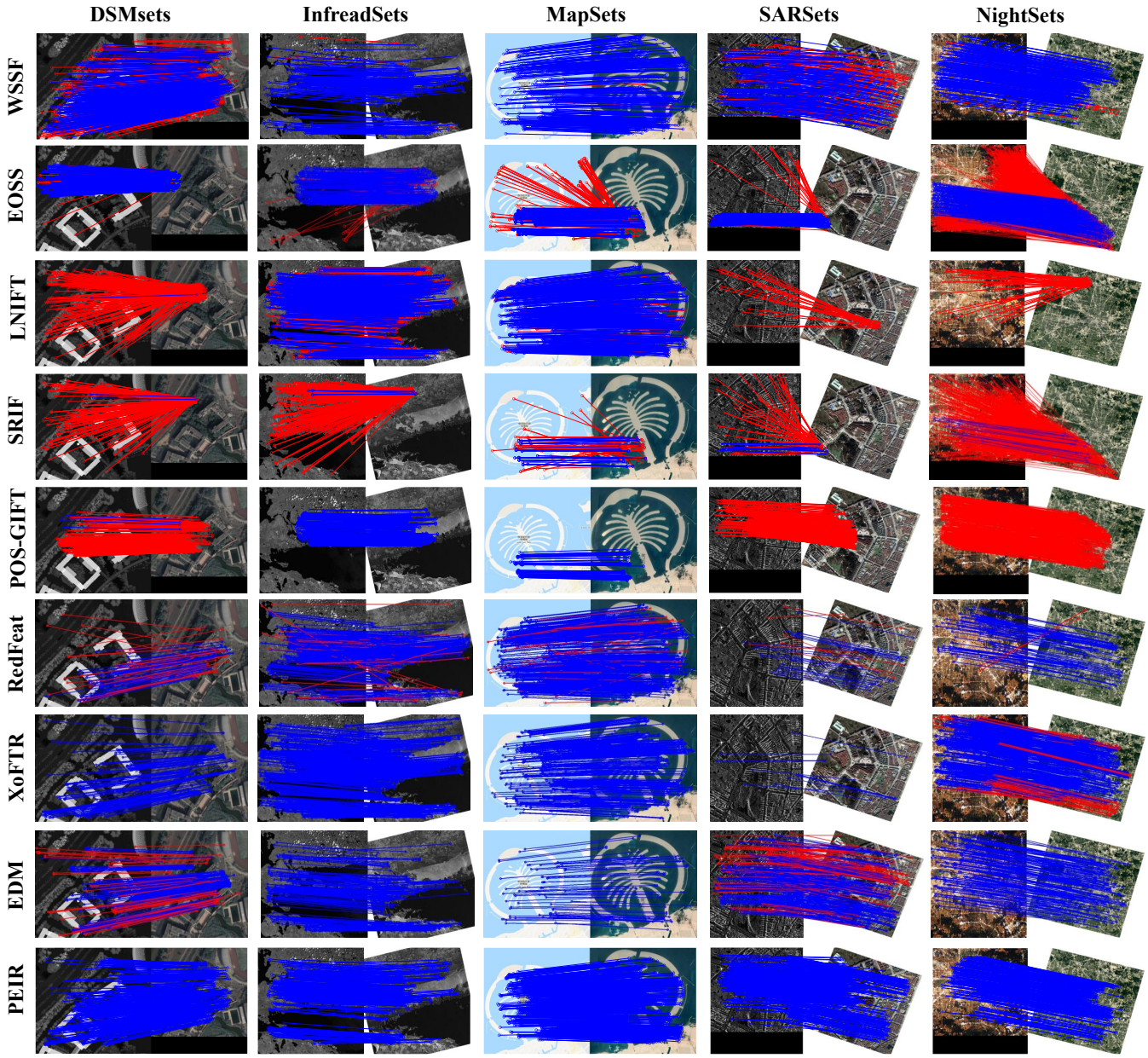

Fig. 8: Qualitative comparison results of all tested methods for five typical image pairs in HOWP dataset. Blue and red lines indicate correct matches and mismatches, respectively.

some cases, they lack the physical priors necessary for consistent cross-modal robustness under heavy noise and geometric distortions. In contrast, PEIR achieves stable and superior performance across all modalities, which has the highest overall SR of 90.00%. PEIR also records the best CMR in four modalities and the lowest MSE values in three modalities. The qualitative results in Figure 8 demonstrate that the matching accuracy of LNIFT, SRIF and POS-GIFT significantly decreases on image pairs with large rotation angles. PEIR produces significantly denser and more uniformly distributed correct matches (blue lines) with markedly fewer outliers (red lines) compared to all baseline methods.

E. Result on the SRIF Dataset

Table III and Figure 9 report the quantitative and qualitative experimental results on the SRIF dataset. A large portion of the images in this dataset involve random rotations ($a\in[0°,90°]$) and random scale variations ($s\in[0.5,2]$), imposing stricter requirements on the geometric invariance of matching algorithms. As shown in Table III, compared with deep learning-based methods, traditional handcrafted methods achieve a higher overall Success Rate (SR) and demonstrate a certain degree of robustness under these challenging conditions. SRIF achieves the highest SR of 95.67%, yet it suffers from an excessively high false-positive rate, particularly in the Nighttime modality, resulting in a large number of incorrect correspondences

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TABLE III: The quantitative comparison results on SRIF datasets. The best and second-best results in each category are highlighted in Bold and Underline, respectively.

Methods	Metrics	Nighttime	Optical-Depth	Optical-Infrared	Optical-Map	Optical-Optical	Optical-SAR
WSSF	NCM	124.6	137.8	29.9	123.0	618.3	74.5
	CMR(%)	34.22	49.68	12.62	3.25	53.68	26.02
	MSE(↓)	5.416	4.715	6.419	5.262	5.397	5.127
	SR(%)	88.33					
	Time(s)	121.90					
EOSS	NCM	550.8	120.9	429.5	309.5	48.7	239.2
	CMR(%)	65.61	17.83	47.24	56.22	49.66	82.66
	MSE(↓)	5.728	6.728	6.058	5.562	6.364	4.818
	SR(%)	76.67					
	Time(s)	4.14					
LNIFT	NCM	965.4	240.5	414.4	662.7	38.3	330.8
	CMR(%)	38.83	16.47	13.88	13.28	8.01	18.59
	MSE(↓)	5.494	5.360	5.822	5.659	6.698	6.111
	SR(%)	82.67					
	Time(s)	45.81					
SRIF	NCM	258.8	100.4	99.7	130.9	85.9	222.2
	CMR(%)	11.24	6.40	4.29	6.29	10.21	10.69
	MSE(↓)	6.397	4.787	5.977	5.276	5.538	4.892
	SR(%)	95.67					
	Time(s)	1.54					
POS-GIFT	NCM	128.6	96.1	35.72	244.0	28.4	25.0
	CMR(%)	93.60	60.85	39.63	67.94	37.88	64.31
	MSE(↓)	3.017	6.290	7.015	5.085	5.232	4.807
	SR(%)	53.00					
	Time(s)	6.59					
XoFTR	NCM	112.4	27.7	23.1	124.9	101.8	12.5
	CMR(%)	64.94	32.41	20.86	96.69	42.05	100.00
	MSE(↓)	4.375	7.181	7.824	2.987	6.762	1.963
	SR(%)	52.00					
	Time(s)	11.46					
EDM	NCM	548.5	18.5	117.0	316.4	1572.0	68.5
	CMR(%)	42.31	9.62	9.66	37.93	55.82	27.22
	MSE(↓)	5.993	9.283	8.457	6.748	6.338	7.072
	SR(%)	33.00					
	Time(s)	2.08					
ReDFeat	NCM	119.2	47.7	5.3	47.7	48.4	5.0
	CMR(%)	55.70	28.63	21.50	79.63	26.36	15.97
	MSE(↓)	5.799	7.896	7.612	3.530	8.189	8.170
	SR(%)	34.00					
	Time(s)	1.69					
PEIR	NCM	116.8	105.2	403.0	290.1	161.4	181.8
	CMR(%)	86.98	83.82	61.12	97.61	64.11	100.00
	MSE(↓)	3.923	3.955	3.918	3.823	4.312	1.463
	SR(%)	92.33					
	Time(s)	8.20					
Number of image pairs		200	200	200	200	200	200

and elevated MSE values (up to 6.397 pixels). Other handcrafted methods, such as WSSF, EOSS, LNIFT, and POS-GIFT, record moderate SR values, but consistently underperform in NCM and CMR across most pairs. Learning-based methods (XoFTR, EDM, and ReDFeat) demonstrate even greater vulnerability to the imposed rotations and scales, with SR values dropping sharply to 52.00%, 33.00%, and 34.00%, respectively. This is because their data-driven representations are unable to capture the invariant geometric structures without training on diverse multimodal datasets. In contrast, PEIR achieves the second-highest SR of 92.33% while ranking first in MSE for four modalities and attains the highest CMR in five modalities. The qualitative visualizations in Figure 9 further illustrate these advantages, with PEIR producing significantly denser, more uniformly distributed correct matches (blue lines) and substantially fewer outliers (red lines) than all baselines, particularly in complex SAR-Optical and optical-infrared pairs.

We measured the average inference time (in seconds) for each method on the SRIF dataset, as shown in Table III. Handcrafted methods (WSSF, EOSS, LNIFT, SRIF, POS-

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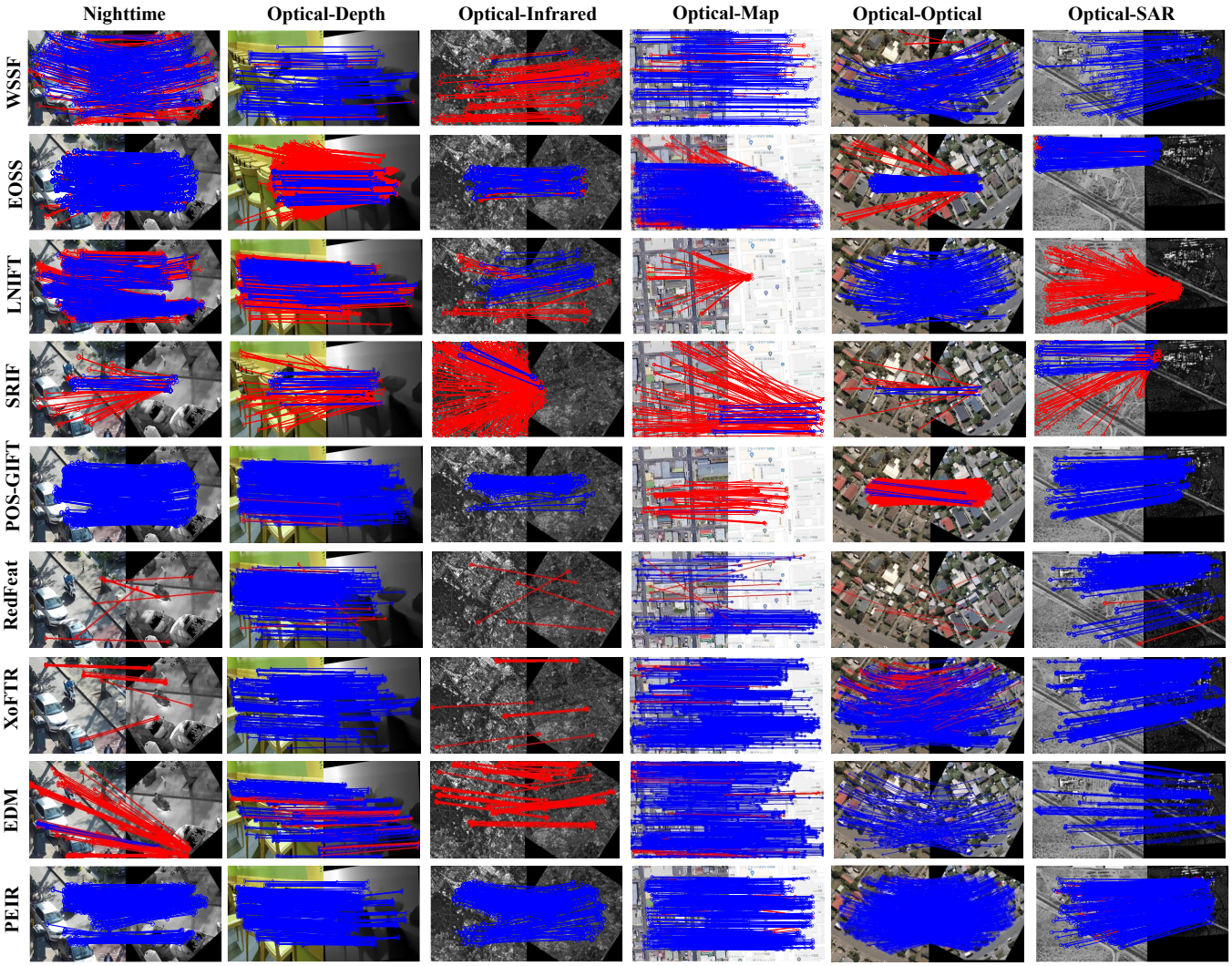

Fig. 9: Qualitative comparison results of all tested methods for six typical image pairs in SRIF dataset. Blue and red lines indicate correct matches and mismatches, respectively.

GIFT, and PEIR) were implemented in MATLAB 2023a and executed on an Intel Core i9-13900HX CPU, whereas learning-based methods (XoFTR, EDM, and ReDFeat) were implemented in Python and executed on an NVIDIA RTX 4090 GPU. Consequently, differences in hardware acceleration, programming environments, software implementations, and optimization prevent a strictly fair runtime comparison between these groups. Comparisons within the handcrafted group are more informative. PEIR provides a good balance between accuracy and efficiency, running comfortably on standard CPUs without specialized hardware, which is advantageous for practical remote sensing applications.

F. Homography Evaluation on Multi-Resolution-SAR-dataset and MapData Dataset

Table IV, Table V and Figure 10 present the homography-based evaluation results on the Multi-Resolution-SAR and MapData datasets. As shown in Table IV, PEIR achieves a clear advantage on the Multi-Resolution-SAR dataset, attaining AUC@5px and AUC@10px values of 53.39% and 77.97%, respectively—significantly outperforming all competing handcrafted and learning-based methods. Although its AUC@3px is marginally lower than that of EOSS, qualitative inspection in Figure 10 reveals that EOSS produces correspondences heavily concentrated in limited regions and accompanied by numerous false matches, indicating instability under severe speckle noise and NRDs. PEIR has the lowest 4-point MSE of 5.68 pixels, demonstrating superior geometric transformation estimation across wide resolution and modality gaps. In contrast, other methods such as WSSF, LNIFT, and EDM exhibit substantially higher 4-point MSE values—289.1, 320.9, and 453.5, respectively—highlighting their limited registration precision. Overall, PEIR exhibits outstanding matching accuracy and robustness on the Multi-Resolution-SAR dataset, particularly excelling in key metrics such as AUC and 4-point MSE. These results validate its significant advantages in complex SAR image registration tasks.

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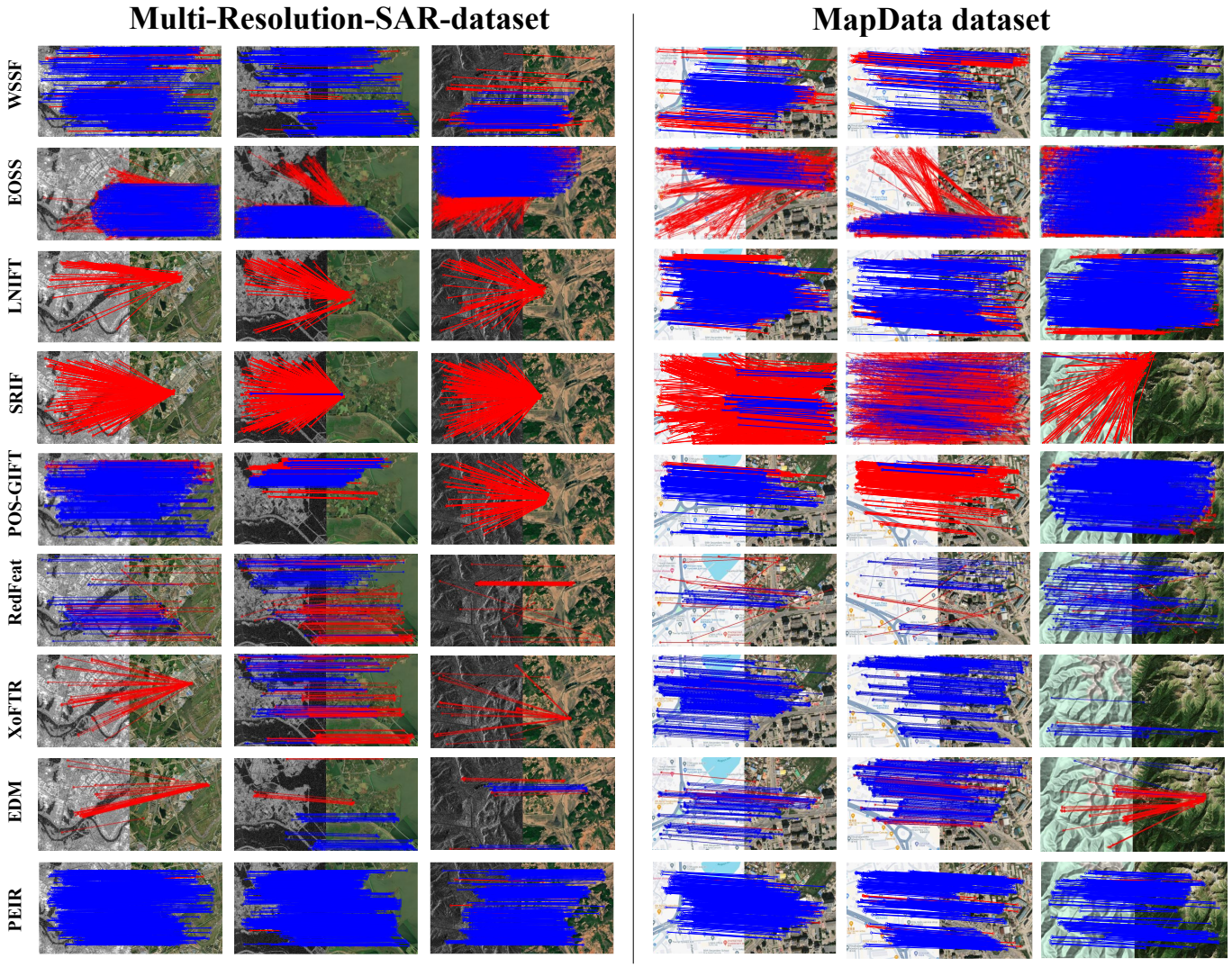

Fig. 10: Qualitative comparison results in Multi-Resolution-SAR-dataset and MapData Dataset. Blue and red lines indicate correct matches and mismatches, respectively.

TABLE IV: The quantitative comparison results on Multi-Resolution-SAR datasets. The best and second-best results in each category are highlighted in Bold and Underline, respectively.

Method	Multi-Resolution-SAR-dataset			
	4-point MSE (↓)	AUC@3px	AUC@5px	AUC@10px
WSSF	289.10	0.85	4.24	5.93
EOSS	19.93	33.90	51.69	75.42
LNIFT	320.90	4.24	9.32	14.41
POS-GIFT	29.22	15.22	42.75	70.29
SRIF	278.80	2.17	5.80	20.29
XoFTR	130.80	25.30	39.60	51.20
EDM	453.50	6.50	11.40	20.50
Redfeat	363.20	15.10	26.60	42.90
PEIR	5.68	27.97	53.39	77.97
Number	1000			

TABLE V: The quantitative comparison results on MapData dataset. The best and second-best results in each category are are highlighted in Bold and Underline, respectively.

Method	MapData-dataset			
	4-point MSE (↓)	AUC@3px	AUC@5px	AUC@10px
WSSF	218.60	5.93	11.86	19.49
EOSS	86.78	45.76	66.10	73.73
LNIFT	54.52	19.49	50.85	76.27
POS-GIFT	30.47	40.68	62.71	77.97
SRIF	188.40	2.90	17.39	35.51
XoFTR	156.10	36.44	46.61	59.32
EDM	404.50	5.93	6.78	8.47
Redfeat	119.30	34.75	51.69	56.78
PEIR	3.06	49.00	69.00	86.00
Number	1910			

As shown in Table V, on the even larger and more geographically diverse MapData dataset, PEIR again delivers state-of-the-art performance across every metric. It has the lowest 4-point MSE of 3.06 pixels and the highest

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TABLE VI: The quantitative comparison results on OFLP datasets. The best and second-best results in each category are highlighted in Bold and Underline, respectively.

Metrics	OFLP	WSSF	FDAFT	HOWP	CoFSM	RIFT	PEIR
NCM	418.3	629.4	795.4	220.4	222.7	112.1	325.9
CMR(%)	27.10	17.75	25.60	13.97	14.63	6.26	25.94
MSE(↓)	1.640	5.130	4.010	6.480	6.180	7.930	3.260
SR(%)	100.00	64.00	77.33	46.67	52.00	26.67	84.00

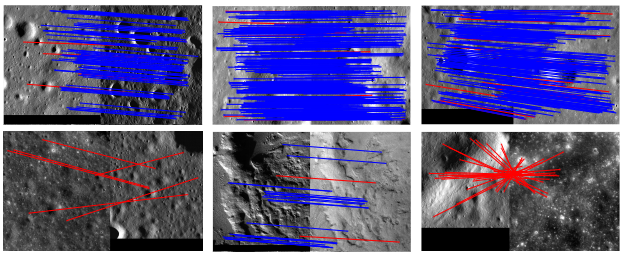

Fig. 11: Qualitative results of PEIR on OFLP datasets. The first line represents successful matching cases; The second row presents failed matching cases.

AUC values (AUC@3 px = 49.00%, AUC@5 px = 69.00%, AUC@10 px = 86.00%). Compared with POS-GIFT, which generates substantially more false correspondences, PEIR produces cleaner match distributions with markedly reduced geometric errors, as visualized in Figure 10. These findings indicate that the PEIR method delivers stable and reliable performance for complex cross-modal registration tasks on the MapData Dataset.

G. Result on the Planetary Remote Sensing Dataset

To evaluate the robustness of our PEIR method in deep-space scenarios characterized by extreme illumination differences, weak textures, and the absence of man-made structures, we conducted experiments on the public planetary dataset provided by Wan et al. (denoted as the OFLP dataset) [58]–[60]. This dataset contains 75 image pairs covering illumination differences, weak textures, and rotation changes, with ground-truth homography matrices.

We compared PEIR with six representative methods: OFLP [58] (a method specifically designed for planetary images using optimized bandpass filtering and local topological priors), WSSF [31], FDAFT [61], HOWP [54], CoFSM [62], and RIFT [14]. All experiments followed the same protocol as in the original OFLP paper, using FSC for outlier removal and the same evaluation metrics: Success Rate (SR), Number of Correct Matches (NCM), Root Mean Square Error (MSE), and Correct Match Ratio (CMR). The quantitative results are shown in Table VI, and the qualitative results are shown in Figure 11.

PEIR achieves a SR of 84.00%, which is significantly higher than WSSF (64.00%), HOWP (46.67%), CoFSM (52.00%), and RIFT (26.67%), and also outperforms FDAFT (77.33%). In terms of MSE, PEIR reaches 3.26 pixels, second only to OFLP (1.64 pixels) and considerably better than all other methods. The CMR of PEIR (25.94%) is competitive, ranking second among all methods. These results demonstrate that PEIR generalizes reasonably well to deep-space scenarios with extreme NRDs and weak textures, although its performance remains below that of the specialized OFLP method.

However, PEIR does not surpass the domain-specific OFLP method. This performance gap may be explained by differences in domain-specific design: (1) OFLP employs a dedicated bandpass filter (B-2D Log-Gabor) that explicitly extracts middle-frequency structures while suppressing both low-frequency contours and high-frequency noise, which is particularly effective for planetary images where true structural information is concentrated in a specific frequency band. In contrast, PEIR uses standard Log-Gabor filters without such bandpass optimization. (2) OFLP introduces a feature block matching strategy guided by local topological consensus, which significantly improves coarse alignment and reduces false matches under extreme illumination variations. PEIR currently relies on point-wise matching with NNDR and FSC, which may be less suited to images with extremely weak structural saliency. These results indicate that the general-purpose PEIR framework does not outperform specialized methods in extreme planetary settings. Future work will focus on integrating frequency-band adaptation (e.g., learnable or adaptive bandpass filtering) and topological consensus mechanisms into the PEIR framework to improve its performance in deep-space scenarios.

H. Rotation Invariance Test

To evaluate the contribution of dominant-orientation assignment to rotation robustness, we compare the complete PEIR with a controlled variant without dominant-orientation assignment on the 2InfraredSets image pairs of the HOWP dataset. The images are rotated from 0° to 360° in 15° increments under the same protocol. In the variant, only the dominant direction estimation is disabled; all other components, parameters, and steps remain unchanged. We record the average NCM and CMR at each angle. As shown in Figure 12, the complete PEIR maintains relatively stable NCM and CMR, whereas removing dominant-orientation assignment leads to lower and more variable matching results, particularly at larger relative rotations. Without dominant direction alignment, corresponding structures can occupy inconsistent spatial and orientation bins in the descriptor; aligning the sampling grid to the estimated dominant direction reduces this sensitivity. This controlled comparison supports the contribution of dominant-orientation assignment to rotation robustness.

I. Hyperparameter setting

PEIR involves three key hyperparameters: the number of scales N_s, the number of orientations N_o, and the relative bandwidth σ_{f} of the Log-Gabor filter. In the main

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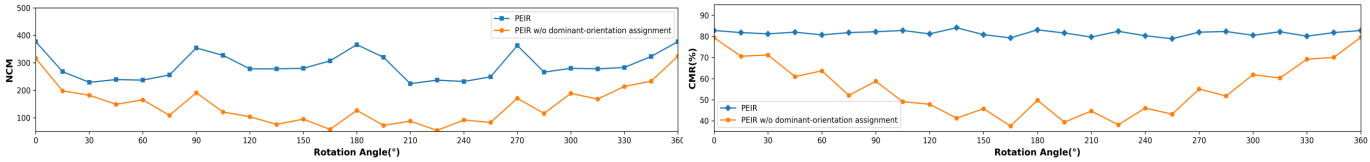

Fig. 12: The rotation invariance test result of PEIR (with and without dominant-orientation assignment), whose abscissa is rotation angle $\theta = [0°, 15°, 30°, ..., 360°]$.

TABLE VII: The results on SRIF subset with different N_s. The best results in each category are highlighted in Bold.

N_s	2	3	4	5	6
NCM	150.2	178.1	224.1	164.8	130.8
CMR(%)	74.04	83.05	94.47	80.48	79.41
MSE(↓)	4.537	4.904	4.338	4.919	4.856
SR(%)	80.00	90.00	96.67	86.67	83.33

TABLE VIII: The results on SRIF subset with different N_o. The best results in each category are highlighted in Bold.

N_o	4	5	6	7	8
NCM	133.9	189.3	224.1	180.9	183.5
CMR(%)	68.41	79.89	94.47	80.73	66.23
MSE(↓)	5.060	4.901	4.338	4.764	4.507
SR(%)	76.67	86.67	96.67	83.33	76.67

experiments, we set $N_s = 4$, $N_o = 6$, and $\sigma_f = 0.75$. To justify this choice, we conducted a sensitivity analysis on a representative subset of the SRIF dataset comprising 10 image pairs from each of the six modalities (60 image pairs in total), covering Nighttime, Optical-Depth, Optical-Infrared, Optical-Map, Optical-Optical, and Optical-SAR modalities. We varied one parameter at a time while fixing the others, and evaluated the matching performance using NCM, CMR, MSE and SR. Table VII shows the test results when varying N_s (with $N_o = 6$, $\sigma_f = 0.75$). Table VIII shows the test results when varying N_o (with $N_s = 4$, $\sigma_f = 0.75$). Table IX shows the test results when varying σ_f (with $N_s = 4$, $N_o = 6$). Based on all results, we confirm that the chosen parameters ($N_s = 4$, $N_o = 6$ and $\sigma_f = 0.75$) provide the best overall balance for PEIR.

TABLE IX: The results on SRIF subset with different σ_f. The best results in each category are highlighted in Bold.

σ_f	0.55	0.65	0.75	0.85	0.95
NCM	206.3	210.0	224.1	140.7	41.9
CMR(%)	86.39	87.78	94.47	83.47	45.24
MSE(↓)	4.841	5.048	4.338	4.874	4.324
SR(%)	90.00	93.33	96.67	90.00	53.33

J. Ablation Studies

1) Component-Level Ablation: To quantitatively validate the contribution of each core component in the proposed PEIR framework, seven controlled variants are evaluated on the SRIF dataset, as shown in Table X. The ablation configurations are as follows: (1) Removing the phase-energy modulated feature detection mechanism in Eq. (5) of Sec. III-B and directly using the original phase consistency map to verify its effectiveness; (2) Replacing the multi-orientation PEIR map in Eq. (2) of Sec. III-A with the original PC map in feature description, to verify its contribution; (3) Removing the Orientation-Context Enhancement (OCE) mechanism from the feature description, to evaluate its impact; (4) Eliminating the dominant orientation assignment method used to achieve rotation invariance, to verify its necessity; (5) Using the classical PC features with the same Log-Gabor parameters ($N_s = 4$, $N_o = 6$, $\sigma_f = 0.75$) to replace PEIR model for both detection and description; (6) Replacing feature saliency map of PEIR algorithm with the maximum moment map of PC calculated using Log-Gabor wavelets; and (7) The complete PEIR algorithm. As illustrated in Table X, the complete PEIR achieves the best overall performance across all evaluation metrics (NCM, CMR, MSE, and SR) compared with the six ablated variants.

Removing the phase-energy modulated detection results in a noticeable degradation in keypoint repeatability, particularly in low-SNR and weak-texture regions (e.g., homogeneous agricultural fields in SAR or shadowed infrared areas), because the model no longer leverages multi-scale energy accumulation to counteract response attenuation inherent in classical PC. Similarly, replacing the PEIR map with the original PC map in the description stage leads to reduced descriptor discriminability and lower CMR, indicating the practical benefit of integrating local response-magnitude information into the descriptor. The absence of OCE leads to higher MSE and lower SR in complex scenes with overlapping directional patterns (e.g., urban intersections or textured boundaries), indicating that incorporating neighboring orientation context contributes to more stable orientation representation for descriptor construction. Eliminating rotation invariance further exacerbates performance drops under the dataset's large geometric transformations, as the descriptor loses alignment with salient structures across modalities. Figure 12 provides more direct evidence of its contribution to rotation robustness. These ablation results rigorously

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TABLE X: The ablation study results on SRIF datasets. The best results in each category are highlighted in Bold.

Methods	Metrics	Nighttime	Optical-Depth	Optical-Infrared	Optical-Map	Optical-Optical	Optical-SAR
w/o phase-energy modulated feature detection mechanism	NCM	99.4	159.4	194.3	315.2	83.6	138.7
	CMR(%)	35.40	14.08	57.92	97.31	34.12	100.00
	MSE(↓)	4.592	4.011	4.052	4.665	4.702	4.199
	SR(%)	89.00					
w/o multi-orientation PEIR map	NCM	35.6	68.4	82.4	164.7	70.9	75.2
	CMR(%)	36.67	15.59	55.96	87.84	38.94	100.00
	MSE(↓)	4.150	4.134	4.664	4.856	4.574	3.983
	SR(%)	77.83					
w/o orientation-context enhancement	NCM	93.6	175.8	250.8	308.3	71.8	140.7
	CMR(%)	31.17	14.42	59.27	96.36	21.35	100.00
	MSE(↓)	4.457	3.966	4.514	4.897	4.381	4.390
	SR(%)	81.50					
w/o dominant orientation assignment	NCM	39.7	68.6	82.6	169.2	61.4	73.8
	CMR(%)	35.39	15.86	55.71	84.09	28.61	95.25
	MSE(↓)	4.488	4.150	4.433	4.749	4.580	5.021
	SR(%)	89.67					
w/o PEIR model (Replace with PC features)	NCM	18.4	44.3	9.0	49.3	25.0	53.2
	CMR(%)	13.20	11.55	7.05	30.97	11.69	39.35
	MSE(↓)	4.333	4.197	5.924	4.400	4.465	3.845
	SR(%)	42.17					
w/o feature saliency map S	NCM	81.3	274.1	27.8	102.8	90.9	34.1
	CMR(%)	48.46	59.99	22.73	50.75	31.77	29.48
	MSE(↓)	4.360	4.074	4.036	4.503	4.811	3.407
	SR(%)	83.33					
PEIR	NCM	116.8	105.2	403.0	290.1	161.4	181.8
	CMR(%)	86.98	83.82	61.12	97.61	64.11	100.00
	MSE(↓)	3.923	3.955	3.918	3.823	4.312	1.463
	SR(%)	92.33					

demonstrate that each module provides complementary and indispensable gains. Replacing PEIR model with classic PC features for detection and description resulted in a significant overall decline in algorithm performance, which confirms that the energy integration and the modulation mechanisms in PEIR provide significant improvements beyond the standard PC. Compared with the maximum moment map of PC, our saliency map S provides superior repeatability and cross modal consistency. In particular, the PC moment map is less effective in some weak-response regions, especially low-SNR SAR-Optical pairs, where it does not incorporate the phase-energy modulation used by PEIR.

2) Response-Level Ablation: To further isolate the contribution of the ternary phase-energy integration in Eq. (2), we conduct an additional response-level controlled ablation study, with the results shown in Table XI. This experiment complements the module-level ablation above: the former specifically isolates the contribution of the ternary response formulation in Eq. (2), whereas the latter evaluates the contribution of the major components of the complete matching framework. Four orientation-specific response representations are compared: the phase-vector magnitude only, the accumulated amplitude only, the complete PEIR formulation, and the classical normalized PC. For the phase-vector magnitude only and accumulated amplitude only variants, we retain respectively the first two components and the third component of Eq. (2). The classical normalized PC is computed using the same Log-Gabor parameters as PEIR. To ensure a controlled comparison, only the low-level response representation is changed, while all other factors remain unchanged for all variants.

As shown in Table XI, the accumulated amplitude-only representation can produce relatively large NCM values, but its CMR remains comparatively low, indicating that response magnitude alone is insufficient to distinguish reliable cross-modal structures from modality-specific high-energy responses. In contrast, the phase-vector-magnitude-only representation provides higher matching precision, benefiting from the cross-scale cancellation of phase-inconsistent responses. Nevertheless, it does not explicitly retain the accumulated response-strength information. The complete PEIR achieves the best overall performance, with an SR of 92.33%, compared with 89.17% for phase-vector magnitude only and 80.83% for accumulated amplitude only. These results indicate that the improvement of Eq. (2) cannot be attributed solely to either the accumulated amplitude term or the phase-vector magnitude; instead, the two components provide complementary information, with the former preserving response strength and the latter reflecting cross-scale phase coherence.

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TABLE XI: Controlled ablation study of different response representations on the SRIF dataset. The best results in each category are highlighted in Bold.

Methods	Metrics	Nighttime	Optical-Depth	Optical-Infrared	Optical-Map	Optical-Optical	Optical-SAR
Classical normalized PC	NCM	18.4	44.3	9.0	49.3	25.0	53.2
	CMR(%)	13.20	11.55	7.05	30.97	11.69	39.35
	MSE(↓)	4.333	4.197	5.924	4.400	4.465	3.845
	SR(%)	42.17					
Accumulated amplitude only	NCM	132.41	119.76	267.83	243.52	139.14	157.69
	CMR(%)	42.77	38.65	35.28	70.46	41.92	63.84
	MSE(↓)	4.68	4.52	4.74	4.36	4.81	3.58
	SR(%)	80.83					
Phase-vector magnitude only	NCM	101.57	98.43	331.26	261.84	146.72	169.35
	CMR(%)	73.49	75.62	55.84	93.27	57.38	94.16
	MSE(↓)	4.08	4.01	4.12	4.02	4.44	2.17
	SR(%)	89.17					
Complete PEIR	NCM	116.80	105.20	403.00	290.10	161.40	181.80
	CMR(%)	86.98	83.82	61.12	97.61	64.11	100.00
	MSE(↓)	3.923	3.955	3.918	3.823	4.312	1.463
	SR(%)	92.33					

TABLE XII: Quantitative analysis of orientation responses before and after OCE on the SRIF subset.

Metric	Before OCE	After OCE
Dominant-to-second-peak ratio	1.29	1.13
Orientation entropy	0.92	0.96
Normalized peak margin	0.07	0.03

3) Analysis of Orientation-Context Enhancement: To further characterize the actual effect of the OCE operation, we analyze the orientation-response distribution before and after OCE on the same representative SRIF subset used in Sec. IV-I. We evaluate the dominant-to-second-peak ratio, orientation entropy, and normalized peak margin, which respectively characterize the separation between the two strongest responses, the dispersion of the orientation-response distribution, and the relative prominence of the dominant response. The results are summarized in Table XII.

As shown in Table XII, the dominant-to-second-peak ratio decreases from 1.29 to 1.13, the normalized peak margin decreases from 0.07 to 0.03, and the orientation entropy increases from 0.92 to 0.96 after OCE. These results indicate that the positive symmetric convolution performs local smoothing and orientation-context aggregation across neighboring channels. Therefore, the role of OCE is local orientation-context regularization that provides a smoother orientation representation for subsequent dominant-orientation index construction.

K. Image registration and fusion

To further demonstrate the practical utility of PEIR for real-world multimodal remote sensing applications, image registration and fusion experiments were conducted on ten representative image pairs selected from the MMIM, HOWP, SRIF, MultiResSAR and MapData datasets, spanning diverse modality combinations, as shown in Figure 13. For each pair, homography transformation parameters were estimated using the correspondences established by PEIR, and the sensed image was warped onto the reference image. Registration results were visualized via checkerboard composites and fusion results were visualized via fusion images, where perfect alignment manifests as seamless edge continuity and the complete absence of ghosting artifacts or color misalignment. These results clearly demonstrate the efficiency and precision of the PEIR method in image registration.

L. Discussion on Overall Performance

Across five diverse multimodal remote sensing datasets—MMIM, HOWP, SRIF, Multi-Resolution-SAR, and MapData—PEIR achieves competitive performance against five handcrafted baselines (WSSF, EOSS, LNIFT, SRIF, POS-GIFT) and three learning-based methods (XoFTR, EDM, ReDFeat), with favorable results on many evaluation metrics. The superior and stable performance of PEIR originates from three tightly integrated innovations that address the limitations of existing approaches. First, the Phase-Energy Integrated Response (PEIR) model combines multi-scale phase congruency with local energy accumulation via Log-Gabor filters (Eq. 2). This design mitigates the response attenuation suffered by classical PC methods in low signal-to-noise ratio or weak-texture areas (e.g., homogeneous SAR fields or shadowed infrared regions), yielding highly repeatable keypoints even under strong NRDs and modality-specific noise. Second, the phase-energy modulated feature detection mechanism (Eqs. 5-6) adaptively enhances weak structural saliency while preserving geometric semantic consistency across modalities, enabling robust keypoint extraction where gradient- or intensity-based detectors fail. Third, the Orientation-Context Enhancement (OCE) mechanism (Eq. 9) aggregates neighboring orientation responses

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Fig. 13: The registration and fusion results of PEIR using 10 pairs of images from test datasets. The first and second columns in each set of images are input image pairs. The third column shows the fusion results. The fourth column shows the registration results, and the fifth column shows a localized zoomed-in view of the registration results.

and provides local orientation-context regularization for descriptor construction, which improves orientation representation stability in complex scenes—particularly those with weak textures or overlapping directional patterns.

In contrast to handcrafted baselines, which rely on simplistic gradient or self-similarity cues and thus suffer from modality inconsistency, PEIR embeds physically grounded priors (phase invariance under Fourier transform and multi-scale energy). Compared with learning-based methods, PEIR requires no training data or domain-specific fine-tuning, avoiding overfitting to seen modalities and delivering superior generalization on large-scale, real-world datasets such as MapData. The qualitative registration and fusion results (Figure 13) further corroborate these advantages, showing visually well-aligned checkerboard composites. Collectively, these factors enable PEIR to achieve not only higher quantitative accuracy but also greater practical reliability for operational multimodal remote sensing pipelines. The framework's consistent excellence across diverse datasets and scenarios underscores its value as a robust, explainable solution for high-precision image matching, laying a solid foundation for downstream applications including change detection, image fusion, pan-sharpening, and 3D reconstruction. Nevertheless, the OFLP experiment reveals an important limitation: on specialized planetary imagery, the success rate (SR) and the mean square error (MSE) of PEIR are inferior to those of OFLP. This gap is consistent with the advantages of OFLP's planetary-specific B-2D Log-Gabor frequency-band filtering and local topological priors, which are not explicitly modeled by PEIR. Thus, domain-specific frequency adaptation and topological constraints remain directions for future improvement.

V. Conclusion

In this paper, we presented PEIR, a novel Phase-Energy Integrated Response framework that addresses the core challenges of multimodal remote sensing image matching (MRSIM), including severe nonlinear radiometric differences (NRDs), complex geometric distortions, speckle noise, and low-contrast regions. By integrating multi-scale phase congruency with local energy responses through Log-Gabor filters, PEIR constructs geometric feature representations to improve robustness to illumination, contrast, and modality-specific degradations. The phase-energy modulated feature detection mechanism adaptively enhances the saliency of weak structures, while the OCE strategy aggregates neighboring orientation responses, improving the stability of orientation representation and descriptor construction. These innovations collectively overcome the response attenuation and instability of classical phase-congruency methods.

Comprehensive experiments across five diverse public multimodal datasets (MMIM, HOWP, SRIF, Multi-Resolution-SAR, and MapData) demonstrate that PEIR outperforms state-of-the-art handcrafted methods (WSSF, EOSS, LNIFT, SRIF, POS-GIFT) and recent learning-based approaches (XoFTR, EDM, ReDFeat). Our method

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achieves competitive performance in key metrics, particularly under challenging conditions such as SAR-optical pairs, large rotations/scales, and multi-resolution gaps. Qualitative results further confirm denser, more uniformly distributed correct correspondences with significantly fewer outliers. An additional OFLP experiment indicates that PEIR retains useful performance on planetary imagery but remains inferior to the domain-specific OFLP method in SR and MSE.

PEIR provides a physically interpretable, computationally efficient solution that bridges the gap between traditional feature-based robustness and modern multimodal requirements, enabling reliable synergistic analysis for applications such as change detection, image fusion, and 3D reconstruction. Future work will explore its seamless integration with deep learning architectures for improved scalability, real-time deployment on edge devices, and extension to additional modalities (e.g., hyperspectral and LiDAR). Overall, this framework advances robust multimodal remote sensing image matching and offers a solid foundation for next-generation photogrammetry and remote sensing systems.

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